

UNIT 1

Introduction

Introduction and definition, Features of a Project, Types of Projects, Benefits and Obstacles in Project Management, Project Management Profession, Role of Project manager, Consultants, Project and Operation, Project Management Process, Project Scope

1.1 Introduction

- A farmer taking up crop cultivation
- A construction company constructing a bridge
- Indian Railways changing the meter gauge railway track to broad gauge
- A company hiring fresh graduates
- A student pursuing MBA

What is common to all these?

All these are projects. A project is not merely establishing an industry or constructing a building. It is just something new, something unique, planned and executed for good.

Objectives of Project management

There are four major objectives of project management

- **Scope:** Scope means what are the expectations from you as a project manager and your team.
- **Performance:** A project is always expected to have a well defined performance level. If a project is unable to adhere to the desired performance of a customer, it is certainly an unsuccessful project.
- **Time:** A successful project is the one which is completed within the time limits perceived during the planning. As the cost is dependent on time, time management becomes a crucial activity of project management.
- **Cost:** It is dependent on all the above objectives. Mathematically it can be written as: $\text{Cost} = f(P, T, S)$. Therefore, cost is a function of performance, time and scope. If any of the above increases, it is surely going to increase the cost of the project.

Another approach in defining the objectives is the SMART approach.

- **Specific:** Project should target a specific goal
 - **Measurable:** It should be quantifiable
 - **Attainable:** It should be attainable with resources available
 - **Realistic:** It should be realistic in nature
 - **Time Limit:** There should be fixed time limits
-

1.2 Meaning of Project

Project is defined as temporary but interrelated tasks undertaken to give a unique product or service or result. Projects are different from other ongoing operations in an organization, because unlike operations, projects have a definite beginning and an end - they have a limited duration. Projects are critical to the realization of performing organization's business strategy because projects are a means by which the strategy of the company is implemented.

A project can be defined as a complex of non-routine activities that must be completed with a set amount of resources and within a set time limit. The following figure explains the basic tenets of project management.



Typical examples of projects include: construction of a house, performing a marriage, overhauling a machine, maintenance of equipment, commissioning of a factory, conducting national elections, research on developing a new technology, launching a new weapon system, conducting a war, pre-crisis planning for preventing a riot, recruitment of a project manager, etc. Each of the above cases involves investment of resources on a package of inter-related, time-bound activities, thereby constituting a project.

Projects also involve one or more elements that have not been done in the past, and are therefore unique. A product or service may be unique even if the category to which it belongs is large. For example, although several residential complexes have been built in the past, creation of a new house will be a project because each facility can have elements such as a unique - location, customized or adapted design, regionally available resources, and/or discrete owners.

1.3 Definition and No Change Mode

PMBOK (Project Management Body of Knowledge) defines project as a temporary endeavour undertaken to create a unique product or service. Temporary means that every project has a definite end, and Unique means that

the product or service is different from all similar products or services.

Following are the important aspects of a project:

- Starting date
- Specific goals and conditions
- Defined responsibilities
- Budget
- Planning
- Fixed end date
- Parties involved

Project Management

Project management is a methodical approach to planning and guiding project processes from start to finish. It is the method of planning the plan. It starts from project definitions and ends with goal achievement.

PMBOK defines project management as the application of knowledge, skill, tool and techniques to project activities in order to meet stakeholder's needs and expectations from a project.

Bridge group defines it as the methods and disciplines used to define goals, plan and monitor tasks and resources, identify and resolve issues, and control costs and budgets for a specific project.

1.4 Features of a Project

The various characteristics of the project are

1. **Fixed set of objectives:** The project starts when the objective(s) is finalized. The project comes to an end as soon as the objectives are attained.
2. **Tenure:** Project is never a continuous activity, it has to come to an end. Its life span is fixed
3. **Team work:** It needs a team to accomplish various activities.
4. **Unique:** All projects are unique in themselves, no two projects are exactly similar.
5. **Life cycle:** Like all living organisms, project starts slowly (definition phase), then starts building up in size (planning phase), then reaches peak

(implementation phase) before finally getting terminated.

6. **Made to order:** The customer always decides the objective and informs the constraints like time and cost.
7. **Single entity:** Generally, projects are the responsibilities of a single person/entity but certainly there are many participants in a project, who are helping the single entity in the accomplishment of project objectives.
8. **Multi-skilled staff:** The staff needed for a project, including the project manager needs to have a wide range of skills including technical skills, human skills, financial skills, negotiation skills, etc.
9. **Subcontracting:** Subcontracting is practically unavoidable in project management. As specialized knowledge or workforce is needed for a very small duration in a project, it is difficult and costly to employ or retain. Therefore, they are just hired for small duration or specific job from outside agency.
10. **Risk and uncertainty:** Projects are risky as the activities involved in projects are non-retrievable. Thus, risk is unavoidable. However, risk can be reduced considerably using various forecasting techniques and project management and control tools.

1.5 Types of Projects

The projects can be classified into various types:

1) Based on Ownership

a) **Public Projects:** These are the projects which are done by public projects. E.g. Construction of Roads & Bridges, Education Programmes, etc.

b) **Private Projects:** These are the projects which are undertaken by private enterprises. Eg. Any business related projects such as a construction of houses by real estate builders, software development, marriage contracts, etc.

c) **Public Private Partnerships:** These projects which are undertaken by both government and private enterprises together. E.g., Generation of Electricity by Windmill, Garbage Collection, etc.

2) Based on Investment

- a) **Large Scale Project:** These projects involve a huge outlay or investments, say, crores. Eg. Real Estate Projects, Road Construction of manufacturing facilities, Satellite sending projects of ISRO, Unique Identification Number project of India, etc.
- b) **Medium Scale Project:** These projects involve medium level investment and are technology oriented. Example: Computer industry and electronic industry.
- c) **Small Scale Project:** These projects involve only a lesser investments. E.g., agricultural projects, manufacturing projects.

3) Based on Research in Academia

- a) **Major Projects:** In academia, the major projects are those projects which involve more than one year to 3 or 5 years and minimum funding of 3 lakhs in case of social sciences and 5 lakh in case of sciences.
- b) **Minor Projects:** The minor projects in academia are those projects which will be completed within a year and have a maximum funding of 1 lakh in social science and 3 lakh in case of sciences

4) Based on Sector

- a) **Agricultural Projects:** These are the projects which are related to agricultural sector like irrigation projects, well digging projects, manuring projects, soil upgrading project, etc.
- b) **Industrial Projects:** These are the projects which are related to the industrial manufacturing sectors like cement industry, steel industry, textile industry, etc. For example, technology transfer project, marketing project, capital issue project like IPO, etc.
- c) **Service Projects:** These are the projects which are related to the services sectors like education, tourism, health, public utilities, etc. For example, adult literacy project, medical camp, general health check up camp, etc

5) Based on Objective

- a) **Commercial Projects:** These projects are undertaken for commercial purpose and return on investment is expected out these projects. For
-

example, Toll roads based on BOLT – Build Own Lease Transfer Model or BOOT – Build Own Operate and Transfer Model, Product Launching project.

b) **Social Projects:** These projects are undertaken for social purposes and welfare of the people is the aim of these projects. These projects are undertaken either by the Government or Service oriented NonGovernmental Organizations. For example, Polio immunization Project, Child Welfare Projects, Adult Literacy Projects, etc.

6) Based on Nature

a) **Conventional Projects:** These projects are traditional projects which do not apply any innovative ideas or technology or method. For example, conventional irrigational projects, handicraft projects, etc.

b) **Innovative Projects:** These projects involve the use of technology, high R&D, development of new products and services. These innovative projects can be further classified into

i) **Technology:** Depending on the level of technological uncertainty at the time of initiation of projects, the projects can be classified into:

Low-Tech projects which rely on the existing and well established base technologies;

Medium-Tech projects which rest mainly on existing base technologies but incorporate some new technology or feature;

High-Tech projects in which most of the technologies employed are new, but existent, having been developed prior to the project's initiation; and

Super HighTech projects which are based primarily on new, not entirely existent technologies.

ii) **Research:** Based on the type of research, projects can be classified into:

Exploratory research projects which may generate novel idea in the domain of knowledge; constructive research projects which are mainly done by many technological corporate to find new or alternative solutions to any particular crisis or problems, eg., renewable energy

research or development of the capacity of optical fiber; and

Empirical research projects are very impressive observational type of research in which testing on real life data or analysis of pattern of some specific events in order to identify the nature or the class of trend that specific phenomenon maintains.

- iii) **New product development:** These projects are undertaken in the life cycle of a product. These projects can be classified into advance development projects which aim at inventing new science or capturing new know-how for the organization; breakthrough development projects which create the first generation of an entirely new product and involve significant change in the product and process technology; platform or next generation development projects which provide a basis for a product and process family and thus establish the basic architecture for follow-on derivative projects; and derivative development projects which refine and improve selected performance dimensions.

7) Based on Time

a) **Long term projects:** These projects take a very long duration to complete. These projects are run for many years till the objective is reached. For example, Eradication of diseases like Polio, Filaria, etc.

b) **Medium term projects:** These projects take a medium term duration like 3 to 5 years. For example, Modernization projects, computerization of operations, etc.

c) **Short term projects:** These projects are executed within a short period, normally within a year. For example, Pond cleaning project, health camps, software development, etc.

d) **Very short term projects:** By very name you can understand that these projects are completed within a very short period, say, within a day. For example, product launch project.

8) Based on Functions

Based on the functional area of management, the projects can be classified into:

a) **Marketing Projects** which are taken up in the area of marketing a product or service of an organization. Marketing road shows, implementing a marketing strategy, etc.

b) **Financial Projects** are undertaken to raise finance or restructure capital structure. For example, IPO Project, share split project, etc.

c) **Human Resources Projects** are undertaken in the area of human resources of an organization, e.g., Induction training project, campus recruitment project, etc.

d) **IT and Technology Projects** which are undertaken in the area of IT companies or IT related requirement of any organization, e.g., development of Human Resources Information System, Marketing Information System, etc.

e) **Production Projects** are undertaken in the area of production or operations. For example, overhauling projects, preventive maintenance projects, getting an ISO certification, etc.

f) **Strategic Projects** are taken by the organizations to execute a strategy, for example, mergers and acquisition projects, Core Banking Solution project introduced in banks, etc

9) Based on Risk

a) **High Risk Projects:** These projects involve a very high degree of risk, for example, nuclear energy project, thermal energy project, satellite projects, etc. If the project is not handled properly, the effect will be very adverse. Thus, high precautionary measures are to be taken to commission these projects.

b) **Low Risk Projects:** These projects do not involve risk and they are carried out in the normal course of action. For example, road and bridge construction, house construction.

10) Based on Investment Decisions

On the basis how the projects influence the investment decision products, project can be classified into

a) **Independent Projects:** An independent project is one, where the acceptance or rejection does not directly eliminate other projects from

consideration or affect the likelihood of their selection. For example, if management plans to introduce a new product line, as well as, replace a machine which is currently producing a different product. These two projects can be considered independent of each other, if there are sufficient resources to adopt both, provided, they meet the firm's investment criteria.

b) **Mutually exclusive Projects:** The mutually exclusive projects are projects that cannot be followed at the same time. The acceptance of one prevents the substitute proposal from accepting. Most of them have 'either or' decisions. You will not be able to follow more than one project at the same time. The evaluation is done on a separate basis so that one that brings the highest value to the company is chosen.

c) **Contingent Projects:** A contingent project is one where the acceptance or rejection depends on the decision to accept or reject multiple numbers of other projects. Such projects may be complementary or substitutes. Let us take the example of bio fuel plant cultivation in a large scale and the decision to set up a bio fuel manufacturing unit. In this case, the projects are complementary to each other. The cash flows of the plant cultivation will be enhanced by the existence of a nearby manufacturing plant. Conversely, the cash flows of the manufacturing unit will be enhanced by the existence of a nearby cultivation farm.

11) Based on Output

Based on output, projects are classified into quantifiable and non quantifiable ones.

a) **Quantifiable projects:** In these projects, the benefits / goals of which are amenable for measurement. Quantitative expression of the outcomes is possible. It is easy to understand and appreciate quantitative projects as it is easy to communicate them. For instance, enterprises engaged in the production of various goods and services come under this category.

b) **Non-quantifiable projects:** In these projects quantification of the benefits / outcome may not always be possible as the impact of the project is spread over a longer period. The benefits accrue to the intended beneficiaries in the long run. Projects concerning health, education, and environment fall under this category

12) Based on Financial Institutions

Classification Financial institutions – both central and state level have classified projects into profit-oriented projects and service-oriented projects.

a) **Profit-Oriented Projects:** They are classified into a) New Projects; b) Expansion Projects or Development projects; c) Modernization Projects or Technology Projects and d) Diversification Projects.

b) **Service-Oriented Projects:** They are classified into a) Welfare Projects; b) Service Projects; c) Research and Development Projects and d) Educational Projects

1.6 Benefits of Project Management

- Minimizing the need for continuous reporting.
- Identification of time limits for scheduling.
- Identification of a methodology for trade-off analysis.
- Measurement of accomplishment against plans.
- Early identification of problems so that corrective action may follow.
- Improved estimating capability for future planning.
- Knowing when objectives cannot be met or will be exceeded.

1.7 Obstacles in Project Management

To enjoy the various benefits of project management given above, the following obstacles be overcome carefully.

- ◆ Project complexities
- ◆ Execution of customer's special requirements
- ◆ Organisation restructuring is a typical task
- ◆ Project risks
- ◆ Changes in technology
- ◆ Forward planning and pricing.

1.8 Project Management A Profession

Project management has been evolved as a distinct ever since the Second World War. It has got elevation the recent times.

Novelty is the hallmark of every project, hence it should exhibit fascination and dynamism. This requires professional approach in conceiving, implementing and controlling projects. Though the functional management and project management are related, the degree of professional approach is highly essential for the efficient management of project. The project management is mainly driven by intellectual operation and skilled and mechanical operations. Project management is covered by the matrix form of organization structure where a roles are defined according to a combination rather than functional specialization.

Only managers with sufficient spirit and dynamism can withstand the over whelmin dizziness in these incessant operations.

Hence, the project management requires sound expertise and exposure, which may not be possessed by the project promoter. So they have to resort the assistance from projects consultants and project managers. A brief description about the role of project manager and need functions of project consultants are given below.

1.9 Project Manager and His Role

A project manager is always found shard in the enternal circle of doing, learnig and changing. Only managers with sufficient spirit and dynamism can with stand the overwhelming dizziness in these incessant operations. An ideal candidate for project managership should have some prominent personal characteristics as out lined by R Archibald.

- Flexible and adaptable
- Preference for significant initiative and leadership
- Aggressiveness, confidence, persuasiveness, verbal fluency;
- Ambition, activity, forcefulness;
- Effectiveness as communicator and integrator;
- Broad scope of personal interests;
- Poised with enthusiasm, in agitation, spontaneity;

Able or willing to devote most of his time to planning and controlling,

- Able to identify problems;
- Willing to make decisions that are acceptable;
- Able to maintain a proper balance in the use of time,

This ideal project manager would probably have doctorates in engineering business and psychology, sustained with a handful years of experience on similar natured project officer occupying different positions, and should have physical fitness to undertake such Machiavellian tasks with feeling of positive stress. Good project managers in industry today would probably be lucky to have 60% to 80% these traits. good project managers are willing to identify their shortcoming and know heavy traffic, they have to balance between the wheels that are mutually exclusive and yet engineering to run coherently, they ensure that goal is reached by properly accelerating the vehicle the vehicle to manager the traffic avoiding.

1.10 Project Consultants

Need of consultant arises:

- i) When a project of new technology is undertaken.
- ii) When the in-house consultant is incapable of meeting the requirements of the project.
- iii) When there is no in-house facility available in the organization.
- iv) When the project is executed on the basis of imported technology and know how.
- v) To avail the advantages of expertise available with the outside consultants.

Consultants may be of:

- a) In-house consultants
- b) Outside consultants
 - Indigenous
 - Foreign consultants.

As regards 'in-house consultant', it may be stated that in many organizations a separate department is maintained in the total organisation structure. This department looks after the work of detailed engineering, drawings and preparation of technical specifications, etc. An office order shall be issued assigning the jobs along with scope of work, time schedule and job responsibilities to be carried out.

While selecting outside consultants the various factors to be considered are : job requirements, facilities available in their organizations, experience, performance, their organization structure, fees, the terms and conditions, pre and post commissioning services etc.

Job of consultants

The functions of a project management consultant have been identified as
1) Assisting the agency in appropriate site investigation and sourcing of materials.
2) Assisting the agency in selecting the appropriate contractor; 3) Checking the quality of work, supervision control, testing monitoring and progress reporting, checking measurements and of bills.

1.11 What is Operation?

Operations are the ongoing execution of activities and they follow an organization's procedures to produce the same result or a repetitive service.

Operations are permanent in nature.

Production, manufacturing, and accounting are examples of operations.

There are many definitions of operations. Some are as follows:

- Operations do not produce new things, but they are necessary to maintain and sustain the system.
- Operations are used to run regular business models, achieve the goals of the business, and support the business.
- Operations are different from projects, which are known for their uniqueness.
- Operations are permanent, and their only constraint is to make a profit for the organization.

Any manufacturing or production process can be an example of an operation.

1.12 Difference between Project and Operation

Project	Operations
Temporary 'organisation' & responsibilities.	Permanent organisation and responsibilities.
Outcome will deliver change & increased benefit to the business. (new or revised organisation's Key Performance Indicators)	Outcome will not deliver change and benefit remains constant.
Defined start and end.	Work is ongoing and repeatable.
Measure progress against timeline & spend against budget.	Measure using organisation's Key Performance Indicators.

1.13 What is Process in Project Management and Process Groups?

Project process corresponds to the project implementation that is divided into sub processes. The structure and process are defined in the project organization considering the attainment of the corporate objectives and therefore also project objectives.

The components of the project process are: project start, project preparation, project planning, project implementation and project completion. The two last components should contain also the project decision. The process is success due to the following methods as personnel, resources, financial resources, establishments, techniques and methods, and present results at the end.

The five PMBOK process groups are:

1. **Initiating Process Group:** Processes required to launch a new project or a new project phase.
2. **Planning Process Group:** Processes related to defining and planning the extent of the project, as well as planning how it will be executed.
3. **Executing Process Group:** Processes related to the actual completion of project activities and tasks.

4. **Monitoring & Controlling Process Group:** Processes covering everything related to tracking, monitoring, reporting on, and controlling project performance and progress.
5. **Closing Process Group:** Processes required to finalize and complete a project or project phase.

1.14 What is Project Scope?

Project scope is a detailed outline of all aspects of a project, including all related activities, resources, timelines, and deliverables, as well as the project's boundaries. A project scope also outlines key stakeholders, processes, assumptions, and constraints, as well as what the project is about, what is included, and what isn't. All of this essential information is documented in a scope statement.

1.15 Difference between Project Group Objectives and Project Scope

Parameter	Objective	Scope
Definition	The objective of a project determines the achievements one intends by completing a project.	The scope of a project includes the limitations one has to face during the project.
Applicability	The objective determines the output for a desired operation.	Scope determines the range of applicability. What cannot be included is out of scope.
Planning phase	During the planning phase, objectives are to be known to make a plan.	During the planning phase, the scope is to be analyzed and according to the scope, the plan has to be prepared to achieve the objectives.
Clarity	Objectives during a process are clear and only if objectives are clear, the process is operational.	Scopes can vary even after a process begins. The scope should be pre-determined but accidental changes are possible.